

Spoken Tutorial – Chain-of-Thought (CoT): Asking the AI to "Think Step-by-Step"

Title: Chain-of-Thought (CoT): Asking the AI to "Think Step-by-Step"

Module	Effective Prompt Engineering
Episode	Chain-of-Thought (CoT)
Learning Objective	At the end of this tutorial learner will be able to 1. By the end of this tutorial, you will be able to use Chain-of-Thought prompting to get better responses from AI models.
Approx. Duration	6 minutes
Outline	• Introduction to Chain-of-Thought (CoT) prompting • Understanding the problem of brief AI responses • What is Chain-of-Thought prompting? • Example 1: Prompt without CoT and its result • Example 2: Prompt with CoT and its detailed result • Why Chain-of-Thought prompting is effective • Other ways to ask for step-by-step thinking • When to use CoT prompting • Summary and Assignment
Meta Tags	Chain-of-Thought, CoT, Prompt Engineering, AI, ChatGPT, LLM, Think Step-by-Step, Detailed Answers, AI Prompting
Pre-requisite Tutorial	Basic understanding of AI chatbots and how to write simple prompts.

Script

Visual Cue	Narration
A welcome screen with 'Chain-of-Thought Prompting' title, gears turning, and a friendly AI icon.	Welcome to this Spoken Tutorial on Chain-of-Thought, or CoT, prompting. Here, we will learn how to ask AI to 'Think Step-by-Step'. This helps AI give more detailed and correct answers.
A lightbulb icon with a checklist, representing learning and objectives.	After this tutorial, you will be able to use Chain-of-Thought prompting. This will help you get detailed and correct answers from AI tools.
Icons for a web browser, an internet connection, and a generic AI chatbot.	To follow this tutorial, you need a web browser like Chrome or Firefox. You also need an active internet connection. Access to an AI chatbot like ChatGPT or Google Gemini will help you practice.
A person thinking with question marks, leading to a simple AI chatbot icon.	Before starting, you should know what AI chatbots are. You should also know how to write simple questions, called prompts, to get basic answers from these tools.

A person looking confused at a computer screen displaying a very short, unhelpful AI response.	Have you ever asked an AI a question and got a very short or confusing answer? Sometimes, the AI gives only one line when you need a detailed explanation. This can be a problem.
A visual representation of a thought process, perhaps a brain icon with arrows showing steps.	This happens because the AI does not understand that it needs to think step-by-step. Chain-of-Thought, or CoT, prompting helps the AI to 'think' before giving the final answer.
A flowchart diagram with 'Problem' -> 'Step 1' -> 'Step 2' -> 'Solution'.	CoT means asking the AI to break down a problem and explain how it got the answer. It is like asking someone to show all steps in a math problem, not just the final answer.
A simple text prompt 'Explain the water cycle' displayed on a chat interface.	Let's understand this with an example. Open your chatbot. Ask the AI: 'Explain the water cycle.' Type this prompt in the message box and submit it.
A chat interface showing the short AI response to 'Explain the water cycle'.	The AI will give a very short answer. For example: 'The water cycle is the continuous movement of water on, above, and below the surface of the Earth.' This answer is correct, but it is not detailed.
A chat interface showing the prompt 'Think step-by-step and explain the water cycle in detail'.	Now, let's add 'Think step-by-step' to our prompt. Ask the AI: 'Think step-by-step and explain the water cycle in detail.' Type this prompt in your chatbot and submit it.
A chat interface showing the start of a detailed AI response, listing stages of the water cycle.	Notice the difference. The AI now gives a more structured and detailed answer. It starts by listing the main stages like Evaporation, Condensation, and Precipitation. Look at this result on your screen.
A chat interface showing the detailed explanation of 'Evaporation' and 'Condensation' from the AI.	The AI then explains each stage clearly. For example, 'Evaporation: The sun heats water in oceans, rivers, and lakes, turning it into vapor.' This makes the explanation easy to understand.
A visual metaphor of a complex puzzle being broken into smaller, solvable pieces.	CoT prompting works because it makes the AI process information in a logical way. It breaks down big tasks into smaller, easy steps. This gives us better quality answers.
A list of alternative phrases for CoT prompting, like a bulleted list on a whiteboard.	You can use other phrases too, not just 'Think step-by-step'. Try 'Walk me through your reasoning', 'Explain your thought process', or 'Break this down for me'. You can try these different options.
Icons representing coding, problem-solving, and learning.	Use CoT when you need detailed explanations or logical thinking. It is very useful for coding, finding errors, or understanding difficult topics.
A summary slide with key takeaways: 'CoT prompting', 'Think step-by-step', 'Detailed answers'.	In this tutorial, we learned about Chain-of-Thought, or CoT, prompting. We saw how adding 'Think step-by-step' helps AI give detailed and logical answers. This makes AI responses better.

A person using a laptop, with a thought bubble showing a car engine and 'Think step-by-step'.	For your assignment, open an AI chatbot. First, ask it: 'Explain how a car engine works' without CoT. Then, ask the same question using 'Think step-by-step'. Compare the two answers and see the difference.
Logos of EduPyramids, SINE, IIT Bombay, with a 'Thank You' message.	This Spoken Tutorial is brought to you by EduPyramids Educational Services Private Limited, SINE, IIT Bombay. Thank you for joining us!